

# Applications of Thales’ theorem

The theorem inside a triangle: proportional segments, the line through a midpoint, and the measurements you cannot reach with a tape.

LESSON 13–14

2 × 40 MIN

GEOMETRY 8, ТЕМЫ 9–10

STAGE 9 · 13.4

LESSON CLOCK

5'

12'

8'

13'

2'

|             |        |
|-------------|--------|
| Warm-up     | 5 min  |
| Explanation | 12 min |
| Interactive | 8 min  |
| Practice    | 13 min |
| Homework    | 2 min  |

LEARNING OBJECTIVES

- Apply Thales’ theorem inside a triangle to find unknown lengths.
- Prove that a line through the midpoint of one side, parallel to a second, bisects the third.
- Divide a segment in a given ratio by construction.
- Solve practical measuring problems using proportional segments.

TEXTBOOK

|           |                            |
|-----------|----------------------------|
| National  | Темы 9–10, pp. 23–28       |
| Cambridge | Learner’s Book pp. 293–299 |
| Workbook  | Workbook 13.4              |

01

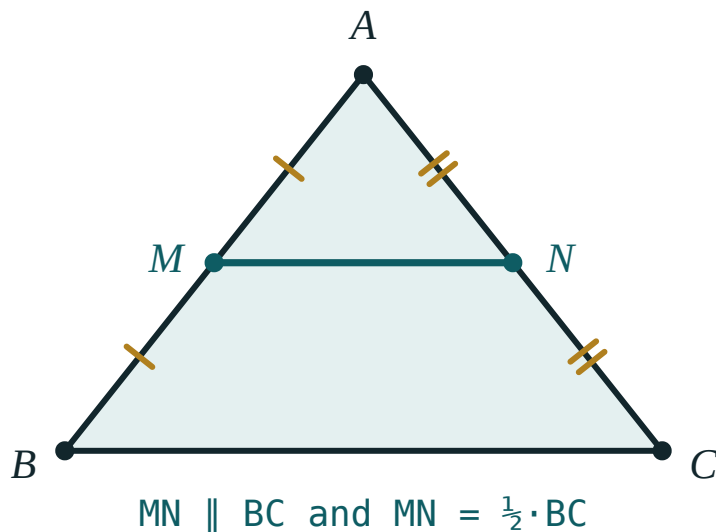
## Explanation

### Thales inside a triangle

THE FORM YOU WILL USE MOST

If a line parallel to  $BC$  cuts  $AB$  at  $M$  and  $AC$  at  $N$ , then

$$\frac{AM}{MB} = \frac{AN}{NC} \quad \text{and} \quad \frac{AM}{AB} = \frac{AN}{AC}$$



*When  $M$  and  $N$  are the midpoints, the ratio is  $1 : 1$  —  
the special case that gives the midline.*

Two different ratios, and mixing them up is the commonest error in this topic. Decide first whether you are comparing a **part with a part** ( $AM : MB$ ) or a **part with the whole** ( $AM : AB$ ), and stay with that choice on both sides.

### The midpoint corollary

#### COROLLARY

A line drawn through the midpoint of one side of a triangle, parallel to a second side, bisects the third side.

**Proof.** With  $AM = MB$  the ratio  $\frac{AM}{MB}$  is 1, so  $\frac{AN}{NC}$  is also 1 and  $AN = NC$ . ■

This corollary is what makes the midline theorem of the next lesson possible, so it is worth stating separately.

### Dividing a segment in a given ratio

To divide  $AB$  in the ratio  $2 : 3$ :

1. Draw a ray from  $A$  and step off  $2 + 3 = 5$  equal segments.
2. Join the 5th point to  $B$ .
3. Through the 2nd point draw a parallel to that line.
4. It meets  $AB$  at the point dividing it in the ratio  $2 : 3$ .

The same construction, with  $n$  equal steps and a parallel through each, divides the segment into  $n$  equal parts.

## Measuring what you cannot reach

The practical use of the theorem is measuring across an obstacle — a river, a building, a fenced field. Set up two similar configurations sharing an angle, measure the three lengths you can reach, and solve the proportion for the fourth.

### WORKED PATTERN

A pole  $1.5\text{ m}$  tall casts a shadow  $2\text{ m}$  long; at the same moment a tree casts a shadow  $14\text{ m}$  long. The sun's rays are parallel, so

$$\frac{1.5}{2} = \frac{h}{14} \implies h = 10.5\text{ m}$$

## 02 Worked examples

**EXAMPLE 1** In  $\triangle ABC$ ,  $MN \parallel BC$  with  $AM = 4$ ,  $MB = 6$  and  $AN = 6$ . Find  $NC$ .

$$\textcircled{1} \quad \frac{AM}{MB} = \frac{AN}{NC}$$

Part with part on both sides.

$$\textcircled{2} \quad \frac{4}{6} = \frac{6}{NC}$$

Substitute.

$$\textcircled{3} \quad 4 \cdot NC = 36$$

Cross-multiply.

$$\textcircled{4} \quad NC = 9$$

### ANSWER

$$NC = 9$$

**EXAMPLE 2** In  $\triangle ABC$ ,  $MN \parallel BC$  with  $AM = 3$ ,  $AB = 12$  and  $AC = 20$ . Find  $AN$ .

$$\textcircled{1} \quad \frac{AM}{AB} = \frac{AN}{AC}$$

This time it is part with whole — note  $AB$ , not  $MB$ .

$$\textcircled{2} \quad \frac{3}{12} = \frac{AN}{20}$$

$$\textcircled{3} \quad 12 \cdot AN = 60$$

$$\textcircled{4} \quad AN = 5$$

ANSWER

$$AN = 5$$

**EXAMPLE 3** A 1.8 m man casts a 2.4 m shadow. A tower casts a 40 m shadow at the same time. Find its height.

$\textcircled{1}$  The sun's rays are parallel, so the two right triangles have proportional sides.  
That is Thales in disguise.

$$\textcircled{2} \quad \frac{1.8}{2.4} = \frac{h}{40}$$

Height over shadow, on both sides.

$$\textcircled{3} \quad 2.4h = 72$$

$$\textcircled{4} \quad h = 30 \text{ m}$$

ANSWER

$$30 \text{ m}$$

### 03 Interactive model

Move the spacing slider so the cuts are unequal, and check that the ratios still match on both rays.

**Which proportion do I write?**

Given as **part and part** on the first side.

① The data give

$AM$  and

$MB$  — two pieces of the same side.

So  
use  
part-  
to-  
part.

②  $\frac{AM}{MB} = \frac{AN}{NC} \Rightarrow \frac{4}{6} = \frac{6}{NC}$

③  $4 \cdot NC = 36$ , so  $NC = 9$

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH           | O'ZBEKCHA                  | РУССКИЙ                  |
|-------------------|----------------------------|--------------------------|
| Proportion        | Proporsiya                 | Пропорция                |
| Cross-multiply    | Krest-nakrest ko'paytirish | Умножить крест-накрест   |
| Midpoint          | O'rta nuqta                | Середина                 |
| Scale             | Masshtab                   | Масштаб                  |
| Shadow            | Soya                       | Тень                     |
| Similar triangles | O'xshash uchburchaklar     | Подобные треугольники    |
| Part to whole     | Qism-butunga nisbat        | Отношение части к целому |
| Part to part      | Qism-qismga nisbat         | Отношение части к части  |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

$MN \parallel BC$  with  $AM = 2$ ,  $MB = 3$ ,  $AN = 4$ . Then  $NC$  is:

$MN \parallel BC$  with  $AM = 2$ ,  $AB = 5$ ,  $AC = 15$ . Then  $AN$  is:

A line through the midpoint of  $AB$  parallel to  $BC$  meets  $AC$  at  $N$ . Then  $N$  is:

A 2 m pole casts a 3 m shadow. A 24 m shadow belongs to an object of height:

16 m

36 m

12 m

48 m

## 06 Practice · 21 problems

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Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS



1.  $MN \parallel BC$ ,  $AM = 2$ ,  $MB = 3$ ,  $AN = 4$ . Find  $NC$ .

ANSWER 6

2.  $MN \parallel BC$ ,  $AM = 5$ ,  $MB = 5$ ,  $AN = 7$ . Find  $NC$ .

ANSWER 7

3.  $MN \parallel BC$ ,  $AM = 3$ ,  $AB = 9$ ,  $AC = 12$ . Find  $AN$ .

ANSWER 4

4. A 2 m pole casts a 3 m shadow; a tree casts 24 m. Find its height.

ANSWER 16 m

5.  $M$  is the midpoint of  $AB$  and  $MN \parallel BC$ . What is  $N$ ?

ANSWER The midpoint of  $AC$ .

6.  $MN \parallel BC$  and  $AM : MB = 1 : 2$ . Find  $AN : NC$ .

ANSWER 1 : 2

7.  $MN \parallel BC$ ,  $AM = 4$ ,  $AN = 4$ ,  $NC = 8$ . Find  $MB$ .

ANSWER 8

Medium · the standard question

7 PROBLEMS

1.  $MN \parallel BC$ ,  $AM = 4$ ,  $MB = 6$ ,  $AN = 6$ . Find  $NC$ .

ANSWER 9

2.  $MN \parallel BC$ ,  $AM = 3$ ,  $AB = 12$ ,  $AC = 20$ . Find  $AN$ .

ANSWER 5

3. A 1.8 m man casts a 2.4 m shadow; a tower casts 40 m. Find its height.

ANSWER 30 m

4.  $MN \parallel BC$ ,  $AN : NC = 3 : 5$  and  $AB = 24$ . Find  $AM$ .

ANSWER 9

5. Describe how to divide a segment in the ratio  $2 : 3$ .

ANSWER 5 equal steps on a ray, join the 5th to  $B$ , draw the parallel through the 2nd.

6.  $MN \parallel BC$ ,  $AM = x$ ,  $MB = 8$ ,  $AN = 6$ ,  $NC = 12$ . Find  $x$ .

ANSWER  $x = 4$

7.  $MN \parallel BC$  and  $MN$  cuts off a triangle with  $AM : AB = 2 : 7$ . Find  $AN : AC$ .

ANSWER  $2 : 7$

### Hard · stretch and reason

7 PROBLEMS

1. Prove that a line through the midpoint of  $AB$  parallel to  $BC$  bisects  $AC$ .

ANSWER  $\frac{AM}{MB} = 1$  forces  $\frac{AN}{NC} = 1$  by Thales, so  $AN = NC$ .

2. In  $\triangle ABC$ ,  $MN \parallel BC$ ,  $AM = x$ ,  $MB = x + 3$ ,  $AN = 4$ ,  $NC = 8$ . Find  $x$ .

ANSWER  $\frac{x}{x+3} = \frac{4}{8} \Rightarrow 2x = x+3 \Rightarrow x = 3$

3. A river is crossed by sighting: two parallel lines cut a base line into 12 m and 18 m, and one side of the river measures 20 m. Find the far distance.

ANSWER  $\frac{12}{18} = \frac{20}{d} \Rightarrow d = 30 \text{ m}$

4.  $MN \parallel BC$  with  $AM : MB = 3 : 4$ . The perimeter of  $\triangle AMN$  is 21. Find the perimeter of  $\triangle ABC$ .

ANSWER Sides scale by  $\frac{3}{7}$ , so the perimeter of  $\triangle ABC$  is 49.

5. Divide a segment into 7 equal parts and explain why no measurement of its length is needed.

ANSWER Seven equal compass steps on a ray plus parallels transfer the equality by Thales; the length of  $AB$  is never used.

6. In trapezium  $ABCD$  ( $AD \parallel BC$ ), the diagonals meet at  $O$ . Prove  $\frac{AO}{OC} = \frac{AD}{BC}$ .

ANSWER The parallels  $AD$  and  $BC$  cut the transversals  $AC$  and  $BD$  proportionally.

7. Three parallels cut one transversal into 4 cm and 6 cm and another into pieces whose sum is 20 cm. Find them.

ANSWER Ratio 2 : 3 of 20 gives 8 cm and 12 cm.

## 07 Homework

### Homework — 6 problems

Geometry 8, Темы 9–10, pp. 23–28. Write the proportion before you substitute.

1.  $MN \parallel BC$ ,  $AM = 6$ ,  $MB = 9$ ,  $AN = 8$ . Find  $NC$ .
2.  $MN \parallel BC$ ,  $AM = 5$ ,  $AB = 15$ ,  $AC = 21$ . Find  $AN$ .
3. A 1.6 m person casts a 2 m shadow; a flagpole casts 15 m. Find its height.
4.  $MN \parallel BC$  and  $AM : MB = 2 : 5$ . Find  $AN : AC$ .
5. Divide a 10 cm segment in the ratio 3 : 2 by construction.
6. In  $\triangle ABC$ ,  $MN \parallel BC$  and the perimeter of  $\triangle AMN$  is half that of  $\triangle ABC$ . Find  $AM : MB$ .